MATH11218
Methods and Basically Specialist Revision

Authored by: Lachlan J. W. Wyer

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Contents

1 Introduction **1**
1.1 Indices: . . . 1
1.2 Poynomials: . . . 1
1.3 Algebraic Functions: . . . 2

2 Sum Notation **3**
2.1 Inequalities: . . . 3
2.2 Partial Fractions: . . . 3

3 Vectors **5**
3.1 Polar Co-ordinates: . . . 5
3.2 Vectors: . . . 5

1 Introduction

1.1 Indices:

Within the expression of a^m, a is considered to be the base with the power m being the index. The reason that this is considered useful is in scientific notation to expression large numbers, or simplify repeated multiplication, fractions, roots or polynomials. E.g. $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6$. With these there are a number of different index laws that goes with it:

1. $a^m \cdot a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{m \cdot n}$
4. $(ab)^m = a^m b^m$
5. $(\frac{a}{b})^m = \frac{a^m}{b^m}$
6. $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
7. $a^0 = 1$

1.2 Poynomials:

A polynomial explains/denotes the degree or curviness that a line is at. Such as a constant can be described as being degree 0, linear degree 1, quadratic degree 2, cubic degree 3 and so on.

Quadratics

With quadratics it is important to understand that it can contain multiple different types of roots. For example, iff: $\Delta = b^2 - 4ac$

- $\Delta > 0$, when it contains 2 real roots, e.g. $(x + a)(x + b)$.
- $\Delta = 0$, when there is 1 real root, e.g. $(x + a)^2$.
- $\Delta < 0$, when no real roots are found, e.g $(x + aj)(x + bj)$.

Polynomials with Degrees Greater than 2

When a polynomial contains a degree that is greater than 2, it roots can be found through a process called polynomial long division.

$$
\begin{array}{r|l}
 & x^2 + x - 12 \\
\hline
x+2 & x^3 + 3x^2 - 10x - 24 \\
 & \underline{-x^3 - 2x^2} \\
 & \quad x^2 - 10x \\
 & \quad \underline{-x^2 - 2x} \\
 & \quad\quad -12x - 24 \\
 & \quad\quad \underline{12x + 24} \\
 & \quad\quad\quad 0
\end{array}
$$

$$
\begin{aligned}
\therefore\ x^3 + 3x^2 - 10x - 24 &= (x+2)(x^2 - x - 12) \\
&= (x+2)(x-4)(x+3) \\
x &= -3,\ -2,\ 4
\end{aligned}
$$

1.3 Algebraic Functions:

Within fractional functions there are two main types:

1. Proper fractions, where the denominator is smaller then the numerator.
2. Improper fractions, where the denominator is less or equal to the numerator.

It is important to note, that the greater, equal to, and less than are only relative to polynomial degree. Some parts to with division of fraction is just the multiplication of the recipricol, $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.

$$
\begin{aligned}
\frac{2}{x+1} + \frac{4}{x+2} &= 0 \\
&= \frac{2(x+2) + 4(x+1)}{(x+1)(x+2)} \\
&= \frac{2x + 4 + 4x + 4}{(x+1)(x+2)} \\
&= \frac{6x + 8}{(x+1)(x+2)} \\
\therefore\ x &\neq -2,\ -1
\end{aligned}
$$

2 Sum Notation

2.1 Inequalities:

$<$ - Less than
$>$ - Greater than
$\leq$ - Less than or equal to
$\geq$ - Greater than or equal to

For example:

$$\begin{aligned} 1 - 2t &< 7 \\ -2t &< 6 \\ t &> -3 \end{aligned}$$

$\frac{x}{y} > 0$ OR $xy > 0$, presents two different cases in the possible solutions that it presents:

Case i - x & $y > 0$

Case ii - x & $y < 0$

$\frac{x}{y} < 0$ OR $xy < 0$, also presents two different cases in the possible solutions that it presents:

Case i - $x < 0$& $y > 0$

Case ii - $x > 0$& $y < 0$

2.2 Partial Fractions:

These are used on improper fractions to complete polynomial long division.

$$\begin{array}{r} x^2 + x + 3 \\ 2x-2\overline{)\;2x^3 \quad + 4x - 1} \\ \underline{-2x^3 + 2x^2} \\ 2x^2 + 4x \\ \underline{-2x^2 + 2x} \\ 6x - 1 \\ \underline{-6x + 6} \\ 5 \end{array}$$

$$2x^3 + 4x - 1 = x^2 + x + 3 + \frac{5}{2x-2}$$

Some equations for the partial fractions, always assuming that the denominator is of a higher degree than the numerator:

1. $\frac{f(x)}{(x+a)(x+b)} = \frac{A}{x+a} + \frac{B}{x+b}$
2. $\frac{g(x)}{(x+c)^n} = \frac{A}{x+c} + \frac{B}{(x+c)^2} + \ldots + \frac{Z}{(x+c)^n}$
3. $\frac{h(x)}{(ax^2+bx+c)(x+d)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x+d}$

(e.g.)

$$\frac{2x^4 - 8x2 + 5x - 2}{x^3 - 4x}$$

$$\begin{array}{r} 2x \\ x^3-4x\overline{)\;2x^4 - 8x^2 + 5x - 2} \\ \underline{-2x^4 + 8x^2} \\ 5x - 2 \end{array}$$

$$\frac{2x^4 - 8x2 + 5x - 2}{x^3 - 4x} = 2x + \frac{5x - 2}{x^3 - 4x}$$

$$\begin{aligned}\frac{5x-2}{x(x^2-4)} &= \frac{A}{x} + \frac{Bx + C}{x^2 - 4} \\ &= \frac{A(x^2-4) + (Bx + C)x}{x(x^2-4)}\end{aligned}$$

$$\begin{aligned}\therefore\ 5x - 2 &= A(x^2 - 4) + (Bx + C)x \\ &= Ax^2 - 4A + Bx^2 + Cx\end{aligned}$$

x^2	x^1	x^0
$0 = A + B$	$5 = C$	$-2 = -4A$
$0 = \frac{1}{2} + B$		$A = \frac{1}{2}$
$B = -\frac{1}{2}$		

3 Vectors

3.1 Polar Co-ordinates:

Here just consider the "x" and "y" co-ordinates of a vector as just the relationship between the angle and hypotenuse.

$$r < \theta$$

Where:

- r is the magnitude
- θ is the angle, usually measured from the x-axis

As such, to find the different aspects of polar coordinates use the following formulas.

$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \arctan(\frac{y}{x})$$

For 3D vectors:

$$r < \theta < \varphi$$

3.2 Vectors:

Remember the scene from Minions, Vector(s) need both direction & and magnitude, $\hat{v} = \frac{\vec{v}}{|v|}$.